# Istiak H. Akib

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## ACADEMIC BACKGROUND

**PhD in Astrophysics** Oct 2024 - Present
Observatoire de Paris, Université Paris Sciences et Lettres (PSL), France
ED-127: Ecole Doctorale Astronomie et Astrophysique d'Ile de France
Expected defense: September 2027

**Master in Space Science and Technology** Sep 2022 - July 2024
Observatoire de Paris, Université Paris Sciences et Lettres (PSL), France
- Grade: 17.4/20 (Très Bien - Highest Honours)
- International Incoming Mobility Grant

**Bachelor of Science in Physics (Advanced Major)** Feb 2019 - Aug 2022
Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea
- GPA: 3.67/4.3 (Percentage: 93%)
- Honors Student, KAIST International Scholarship

## SKILLS

**Programming:** Python (Astropy, Pandas, GalPy, Plotly, PyTorch), R, C, ADQL
**Relevant Softwares:** TOPCAT, AstroImageJ, MATLAB, Origin
**Operating Systems:** Linux, Windows
**HPC Cluster Usage:** MesoPSL
**Language:** English (C2), French (B1), Korean (A2), Bangla (Native)

## RESEARCH EXPERIENCE

**PhD Project** Oct 2024 - Present
LIRA (Laboratoire d'Instrumentation et de Recherche en Astrophysique), Observatoire de Paris, France
**Title**: Reevaluation of the Dynamical Masses of Local Group Galaxies
**Supervisor**: Dr. François Hammer, Dr. Yanbin Yang
- Modeling the Milky Way (MW) - Gaia Enceladus/Sausage (GSE) merger to reproduce observational properties of the MW
- Analyzing validity of Globular Clusters as tracers of merger
- Adjusting the Andromeda (M31) merger model to fit the observed rotation curve

**Galaxy Rotation Curve** May 2024 - Sept 2026
GEPI (Galaxies, Stars, Physics and Instrumentation), Observatoire de Paris, France
- Analysis of the MW rotation curve, MW-like galaxies in the cosmological simulations, and the M31 rotation curve from observations and merger modeling
- Rotation curve fitting with different dark matter distribution models

**Local Group Timing Argument** Jan 2024 - Feb 2025
GEPI (Galaxies, Stars, Physics and Instrumentation), Observatoire de Paris, France

- Re-calculating the local group mass using the timing argument by taking into account the major merger at the M31
- Determining the impact of merger histories on the timing argument mass

**Origin of the Dwarf Galaxies in the MW** Sept 2022 - Dec 2024
GEPI (Galaxies, Stars, Physics and Instrumentation), Observatoire de Paris, France

- Calculated orbits of MW Dwarf Galaxies and Globular Clusters under different MW mass models using GAIA EDR3 data
- Investigation of high energy VPOS dwarfs to be originated from a recent M31 major merger tail models under different MW potential and M31 proper motions
- Found 6D association between tidal tail models and LMC related dwarfs. First indication of matter exchange between M31 and MW and explanation for VPOS.

## PUBLICATIONS

1. I. Akib, F. Hammer, Y. B. Yang, The accretion history of the Milky Way V. The kinematics of most globular clusters trace the merger epochs, *A&A*, 710, A349 (2026).
2. I. Akib, F. Hammer, Y. B. Yang, Impact of merger histories on the timing argument estimate of the Local Group mass, *A&A*, 695, L5 (2025).
3. I. Akib, F. Hammer, Y. B. Yang, M. S. Pawlowski, J. L. Wang, An intriguing coincidence between the majority of vast polar structure dwarfs and a recent major merger at the M31 position, *A&A*, 694, A66 (2025).
4. Y. J. Jiao, F. Hammer, I. Akib, Y. B. Yang, H. F. Wang, A Self-Consistent Jeans Analysis of the Milky Way Rotation Curve, *ApJ*, in press (2026).
5. F. Hammer, Y. B. Yang, P. Amram, L. Chemin, G. A. Mamon, J. L. Wang, I. Akib, Y. J. Jiao, H. F. Wang, Dark matter fraction derived from the M31 rotation curve, *A&A*, 694, A16 (2025).
6. F. Hammer, Y. J. Jiao, G. A. Mamon, Y. B. Yang, I. Akib, P. Amram, H. F. Wang, J. L. Wang, L. Chemin, The Milky Way accretion history compared to cosmological simulations, *A&A*, 692, L1 (2024).
7. F. Hammer, Y. B. Yang, I. Akib, The conundrum of the Large Magellanic Cloud mass. How the Clouds impact the Galactic halo, in peer review at *A&A*, 2026

## EDUCATION AND OUTREACH

- Supervising research of 5 Masters and 4 bachelors students 2024-Present
- Organizing LIRA Journal club 2025-2026
- Student Ambassador and Student Buddy at Université PSL 2023-Present
- Tutor for General Physics and Physics lab at the KAIST 2020-2022
- Freshman mentor at KAIST and the KAIST Physics department 2020-2022
- Writer for physics and astronomy at a science magazine 2017- Present

## REFERENCES

**Dr. François Hammer**
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**Dr. Gary A. Mamon**
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**Dr. Marcel Pawlowski**
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